Circulation & health

Move blood

1 Double circulation

Body → vena cava → right atrium/ventricle → pulmonary artery → lungs.

Lungs → pulmonary vein → left atrium/ventricle → aorta → body.

Left ventricle thicker; valves prevent backflow; coronary arteries supply heart muscle.

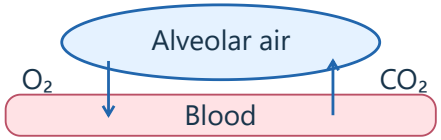
2 Heart rate & flow

Pacemaker cells in right atrium set the rhythm; artificial pacemakers correct irregularities.

Output = rate × stroke volume. 70 × 75 = 5,250 cm^3/min = 5.25 dm^3/min.

Exchange & defend

3 Vessels & lungs



O_2: air → blood; CO_2: reverse

Arteries: thick elastic/muscular walls for high pressure. Veins: wide lumen and valves.

Capillaries: one-cell-thick walls. Alveoli: large area, short path, blood/ventilation maintain gradients.

O_2: alveoli → blood; CO_2: reverse.

4 Blood components

Red cells: haemoglobin, biconcave shape, no nucleus → O_2 transport.

White cells: phagocytosis, antibodies/antitoxins. Platelets: clotting.

Plasma carries cells, nutrients, hormones, CO_2 and urea; distributes heat.

Evaluate disease

5 Disease & treatment

Fat narrows coronary arteries → less O_2/respiration. Stent opens; statin lowers cholesterol.

Faulty valves leak/restrict flow. Compare valve replacement, donor and artificial hearts: benefit, surgery, clot/rejection risks.

6 Health & cancer

Health = physical/mental well-being; diseases interact. Lifestyle/environment/genetics affect risk.

Cancer: uncontrolled division. **Benign:** local; **malignant:** invasion/secondary tumours.

Correlation alone ≠ causation. Compare representative rates and social/financial costs.